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| **Fiscal Year:** | FY 2025 | **Task Last Updated:** | FY 03/21/2025 |
| **PI Name:** | Hayman née Anderson, Allison Ph.D. | | |
| **Project Title:** | Multi-Environment Virtual Training for Long Duration Exploration Missions | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H) | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | allison.p.anderson@colorado.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 417-388-0621 |
| **Organization Name:** | University of Colorado | | |
| **PI Address 1:** | Ann and H.J. Smead Aerospace Engineering Sciences | | |
| **PI Address 2:** | 429 UCB | | |
| **PI Web Page:** | | | |
| **City:** | Boulder | **State:** | CO |
| **Zip Code:** | 80309-5004 | **Congressional District:** | 2 |
| **Comments:** | NOTE: name change to Hayman née Anderson (Ed., March 2025). PI moved to University of Colorado from Dartmouth College in early 2017. | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2020 HERO 80JSC019N0001-HFBP, OMNIBUS3 Crew Health: Human Factors and Behavioral Performance-Appendix E; Omnibus3-Appendix F |
| **Start Date:** | 05/19/2021 | **End Date:** | 09/30/2025 |
| **No. of Post Docs:** | 3 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 3 | **No. of Master' Degrees:** | 2 |
| **No. of Master's Candidates:** | 3 | **No. of Bachelor's Degrees:** | 1 |
| **No. of Bachelor's Candidates:** | 7 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Whitmire, Alexandra | **Contact Phone:** | |
| **Contact Email:** | alexandra.m.whitmire@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date has changed to 09/30/2025 per A. Beitman/HFBP (Ed., 8/14/25).<br>NOTE: End date has changed to 05/18/2028 per NSSC (Ed., 3/25/25). | | |
| **Key Personnel Changes/Previous PI:** | We have added Dr. Lee Miller to the research team as a Co-Investigator (Co-I). He is located at University of California Davis (UC Davis) and is an expert in neurologic measure analysis, specifically EEG. | | |
| **COI Name (Institution):** | Clark, Torin Ph.D. ( University of Colorado, Boulder )<br>Robinson, Stephen Ph.D. ( University of California, Davis )<br>Vance, Eric Ph.D. ( University of Colorado, Boulder )<br>Szafir, Daniel ( University of North Carolina )<br>Miller, Lee ( University of California, Davis ) | | |
| **Grant/Contract No.:** | 80NSSC21K1140 | | |

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| **Performance Goal No.:** | |
| **Performance Goal Text:** | |
| **Task Description:** | Historically, NASA has led the way in virtual reality (VR) technology development to train astronauts for microgravity environments, but there remain open questions regarding the use of adaptive VR for training in long duration exploration missions (LDEM). These missions are likely to have long time periods between pre-mission training and execution of a mission task. Crews will need to train and maintain skills on tasks critical to mission success, such as entry decent and landing (EDL), habitat maintenance and repair, and planetary surface extravehicular activity (EVA). To be successful, performance on these tasks will need to be maintained after high-fidelity training on Earth prior to the start of the mission, and refreshed during transit prior to execution on the surface. The objective of this research is to develop an effective training environment for mission critical LDEM tasks. We will experimentally investigate how skills transfer from the training environment to a physical setting, the degree to which the training environment enables skills maintenance over longer durations, and the generalizability of skills learned in the training environment when transferring to un-trained tasks. As a secondary benefit, we will investigate training in an adaptive, complex VR environment as a countermeasure to the neural decrements demonstrated in spaceflight. Thus, this research will quantify neurophysiological response in the cerebral cortex during training to assess the potential benefit of the trainer as a countermeasure. The novel contributions of this research are to investigate the use of immersive, adaptive VR environments to train astronauts specifically designed for LDEMs by 1) Incorporating multiple, mission critical tasks for a mission to Mars; 2) Investigating skills transfer, retention, and generalizability; 3) Designing with low volume, mass, and operational overhead; and 4) Investigating the degree to which the proposed training can also provide secondary benefits of broad brain stimulation. Our proposed research is directly aligned with that of the Virtual NASA Specialized Center of Research (VNSCOR), and can be integrated with researchers investigating teams, optimal adaptive control algorithms, motivational strategies, or neural and cognitive decrements with both human and animal studies. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This research has wide application in training in immersive environments, particularly for complex tasks. We develop a novel training algorithm to facilitate rapid skills acquisition, while also ensuring broad improvement in performance (rather than unidimensional performance). To our knowledge is absent from the literature, but may improve how we administer training in an automated (rather than supervised) training paradigm. This research will also contribute to our understanding of immersive environments and the degree to which they stimulate the brain. This informs our scientific understanding of neuroplasticity when learning and acquiring new skills. This has applications in rehabilitation, to inform how using these environments may yield better patient outcomes. |
| **Task Progress:** | In the reporting year, we have made substantial progress on all four of our Aims. Below are detailed results as outlined by Aim.<br>Specific Aim 1: Develop a Long Duration Exploration Mission (LDEM)-specific, multi-environment virtual training simulator<br><br>Within the reporting year, we have made progress on our habitat maintenance and repair task. We have completed integrating the 3 subtasks that have been reported previously. We have also performed several rounds of pilot studies to fine-tune the algorithm. We are nearing the point of doing experimental validation of the habitat environment. We have also prepared a journal manuscript for our previously presented study performed in conjunction with the United States Air Force Academy. We have also completed the validation and testing of our surface extravehicular activity (sEVA) environment. We used that environment to perform a human-subject experiment comparing head-mounted displays presented through a virtual reality (VR) headset and 2-Dimensional (2D) screen-based virtual environment presentations for training. Our work in prior years showed improved learning in VR environments with our adaptive difficulty algorithms, compared to a VR display with static difficulty algorithms and 2D presentations with static difficulty algorithms. Thus, these results allow us to elucidate the differences between immersive VR presentation compared to 2D display, informing whether the previous results were primarily due to the adaptive difficulty algorithm we developed. We currently have a journal paper in draft undergoing internal edits.<br><br>Specific Aim 2: Quantify brain activation during training to assess the environment’s efficacy as a countermeasure to neural decrements in spaceflight<br><br>In the reporting year, we were able to complete Aim 2. We currently have a journal paper ready for submission. These results found that individuals had statistically significant increase in brain activation, as measured through electroencephalography (EEG) and functional near-infrared spectroscopy (fNIRS), when training in the complex, operationally relevant task in comparison to the corollary cognitive task. The results were consistent across different regions of the brain. Currently, we are opportunistically analyzing the data in an additional investigation to determine if there are neural predictors of performance during training and task execution. That effort is just beginning and will be reported on in the upcoming year.<br><br>Specific Aim 3: Investigate use of the virtual training environment for performance maintenance<br><br>The primary experiments of this Aim have been completed and previously reported. Two journal publications reporting these findings are in preparation. However, there are two ongoing efforts still associated with this Aim. In the reporting period, we completed data collection with the Polaris Dawn crew, who performed their space mission in January. Initial results demonstrate a decrease in performance immediately post-mission, but data analysis is ongoing. The ground-based cohort has been recruited and trained. They will perform a similar data collection after they have undergone operational stressors as part of the SpaceX ground-support team to elucidate the difference between spaceflight-specific adaptations upon reentry and operational and time-based performance decrements.<br><br>The other remaining effort as part of this Aim is being performed primarily by UC Davis. The team is using the corollary tasks, as described in previous reports, to analyze performance, learning rate, retention, and neural activities between pilots and non-pilots. As this also incorporates neural response, this experiment combines elements from Aims 2 and 3. These corollary tasks have been tested at CU Boulder amongst their student body. However, since these tasks are for an astronaut training algorithm, it is important to know how groups with a similar skillset to astronauts, like aircraft pilots, perform in these tasks compared to those that do not have the applicable experience. We hypothesize that pilots will |

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| | receive better scores and achieve higher accuracy in tasks than non-pilots, learn to reach expertise faster than non-pilots, have better retention than non-pilots, and that there will be a measurable difference between the two groups. We also expect that the corollary tasks done at UC Davis will activate the same parts of the brain as subjects in CU Boulder. Neural activity is measured with EEG, and data collection is ongoing.<br><br>Specific Aim 4: Demonstrate long-term acceptability, engagement, and skills transfer in an isolated, confined, controlled (ICC) environment<br><br>A major recent effort in the reporting period has been to prepare our experiment for the participation in the NASA Human Exploration Research Analog (HERA) Campaign 8 mission. A key aspect of our integration has also been coordinating with the other MATRIKS researchers to ensure alignment of the protocols. The objective of this research is to demonstrate: 1) long-term acceptance and acceptability; 2) assess changes in mood as a result of using the training environment; and 3) skills transfer from Trinity to a novel task in an isolated, confined, controlled (ICC) environment replicating a human space mission. [Ed. Note: NASA has established a Virtual NASA Specialized Center of Research (VNSCOR) referred to as the “Mars Adaptive Training Integrative Knowledge System (MATRIKS)”, which is comprised of the following projects: (1) “Trinity – Multi-Environment Virtual Training for Long Duration Exploration Missions”, (PI: A. Hayman nee Anderson, University of Colorado Boulder); (2) “Morpheus – A Haptic Sensory Supplement to Optimize In-Flight Adaptive Training for Human Control of Spacecraft Robotic Arms” (PI: S. Robinson, University of California Davis); and “Neo – Adaptive Training integrative knowledge System to Improve Operational Performance and its Neural Basis for Spaceflight” (PI: C.W. Jones, University of Pennsylvania).] |
| **Bibliography Type:** | Description: (Last Updated: 02/18/2026) |
| **Abstracts for Journals and Proceedings** | Hayman A, Bonarrigo L, Sherman S, Putman E, Baldonado K, Dutta P, Robinson S, Vance E, Szafir D, Clark T. "Trinity: Multi-environment virtual trainer for long-duration exploration missions." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025.<br>Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 |
| **Abstracts for Journals and Proceedings** | Dutta, P., Clark, T., Hayman, A. "Virtual Reality Training Environment for Space Habitat Maintenance Tasks" NASA Human Research Program Investigators Workshop. Galveston, TX, January 2025<br>NASA Human Research Program Investigators Workshop. Galveston, TX, January 2025 , Jan-2025 |
| **Awards** | Bonnarigo L. "Outstanding Research Mentor, May 2024." May-2024 |
| **Awards** | Sherman S. "AAAS Science Technology Policy Fellow, August 2024." Aug-2024 |
| **Awards** | Clark T. "Elected Fellow, Aerospace Human Factors Association (AsHFA), May 2024." May-2024 |
| **Awards** | Clark T. "Outstanding Graduate Teaching & Mentoring, Ann and H.J. Smead Aerospace Engineering Sciences Department, May 2024." May-2024 |
| **Awards** | Hayman A. "Selected, National Research Council Research Associates Program, January 2025." Jan-2025 |
| **Awards** | Hayman A. "Outstanding Mentor Award, University of Colorado Boulder Summer Program for Undergraduate Research (SPUR) Program, August 2024." Aug-2024 |
| **Dissertations and Theses** | Putman E. "Adaptive training with virtual reality for the maintenance of mission-critical skills in long duration spaceflight." Dissertation, University of Colorado, May 2024. , May-2024 |